# North Pharmacy

## Software Test Plan & Test Coverage Report

Customer Website & Admin Dashboard

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|---|---|
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## Coverage at a Glance

| 14 | 13 | 269 | 4 | 9 |
|---|---|---|---|---|
| Customer Modules | Admin Modules | Test Cases | Weeks Duration | Test Types |

> **Document set.** This plan is implemented by two companion documents: **NP-QA-TS-001** (Test Scenario Report, 106 scenarios) and **NP-QA-TC-001** (Test Case Document, 269 executable test cases with severity and priority). The per-module counts in Sections 7 and 8 are the counts actually designed in NP-QA-TC-001.

# 1. Introduction and Objectives

This document defines the test plan for the **North Pharmacy E-commerce** web platform, covering the two in-scope applications: the public **Customer Website** and the role-protected **Admin Dashboard**. It establishes what will be tested, how it will be tested, by whom, under what conditions, and the criteria by which the release will be judged ready for production.

## 1.1 Objectives

- Verify that every customer journey — discovery, search, cart, prescription upload, checkout and order tracking — works correctly and consistently.
- Verify that every administrative function — order lifecycle, prescription review, catalogue, inventory, reporting, content and store settings — behaves correctly, and that data written by an administrator is reflected accurately on the customer site.
- Confirm the integrity of pricing, discount, delivery-charge and order-total calculations, as these directly affect revenue and customer trust.
- Confirm that prescription-controlled medicines cannot be purchased without a pharmacist-approved prescription.
- Confirm role-based access control: no customer-level or unauthenticated session may reach admin functionality or admin data.
- Validate compatibility across the supported browser and device matrix, including mobile, which is the dominant channel in this market.
- Identify, document and track defects to closure, and provide management with an objective, evidence-based release-readiness assessment.

## 1.2 Reference Documents

| Reference | Description |
|---|---|
| Business Requirement Specification | Functional scope and business rules for the pharmacy platform |
| UI/UX Design Files | Approved screen designs for the customer and admin interfaces |
| API Contract | REST endpoints under `/api/*` consumed by the single-page front-end |
| Regulatory Guidance (DGDA) | Prescription-medicine handling obligations applicable in Bangladesh |
| IEEE 829 / ISTQB | Test documentation structure and terminology followed in this plan |

# 2. Scope of Testing

## 2.1 In Scope

| Application | Included Functional Areas |
|---|---|
| **Customer Website** | Home page and merchandising, department and category browsing, product search with autocomplete, product detail, cart, wishlist, registration and login, password recovery, profile and address book, prescription upload, checkout and payment selection, order placement, order history and tracking, newsletter subscription, static content pages, manufacturer and brand pages. |
| **Admin Dashboard** | Admin authentication and role gate, dashboard KPIs and analytics, order management and status lifecycle, prescription review and approval, product catalogue management, inventory and stock control, manufacturer management, customer management, promotions, reporting suite, website content management, and store settings (business details, delivery, payments, social links). |
| **Cross-Cutting** | Authentication and session handling, authorisation boundaries, input validation, calculation accuracy, admin-to-customer data propagation, responsive layout, browser compatibility, baseline performance, application-level security checks, and regression. |

## 2.2 Out of Scope

- Source-code-level unit testing and code-coverage measurement (developer responsibility).
- Server, network, database and operating-system infrastructure testing.
- Third-party payment gateway certification (bKash / Nagad live settlement). Only the application-side configuration and selection behaviour is tested, since these channels are currently disabled.
- Penetration testing and formal security audit by a certified external party.
- SMS and email gateway deliverability, and carrier-level delivery guarantees.
- Point-of-Sale (POS) and any back-office system outside the two in-scope web applications.
- Load testing beyond the agreed baseline concurrency defined in Section 10.

## 2.3 Assumptions

- The test environment remains stable and available for the full test window, with no unannounced deployments during execution cycles.
- Test data (products, customers, orders, prescriptions) can be created and reset without affecting production operations.
- Administrator credentials with full privileges are provided for the duration of testing.
- Requirements are frozen at the start of Cycle 1; any change after that point is handled through change control and may affect the schedule.

# 3. Application Under Test — Architecture Overview

The platform is a single-page application (React / Vite build) served over HTTPS and communicating with a REST API on the same origin. Both the Customer Website and the Admin Dashboard are delivered from the same front-end bundle; the admin area is code-split and loaded only for authorised sessions. Authentication is cookie/session based with a `rememberMe` option, and all API traffic is sent same-origin.

## 3.1 Application Map

| Area | Front-End Routes | Primary API Surface |
|---|---|---|
| Storefront & Catalogue | `/`, `/shop`, `/manufacturers`, `/page/:slug`, `/terms` | `/api/catalog/home`, `/catalog/products`, `/catalog/categories`, `/catalog/manufacturers`, `/catalog/suggest`, `/store/config`, `/store/content` |
| Authentication | `/login`, `/register`, `/forgot-password` | `/api/auth/login`, `/register`, `/logout`, `/me`, `/forgot/channels`, `/forgot/send`, `/forgot/reset` |
| Cart & Checkout | `/cart`, `/checkout`, `/order-success/:id` | `/api/cart`, `/cart/items/:id`, `/cart/quote`, `/cart/merge`, `/checkout`, `/checkout/options` |
| Customer Account | `/account`, `/account/orders`, `/account/addresses`, `/account/prescriptions`, `/account/wishlist` | `/api/account/profile`, `/account/password`, `/account/addresses`, `/account/summary`, `/api/orders`, `/api/prescriptions`, `/api/wishlist` |
| Admin Dashboard | `/admin/dashboard`, `/admin/orders`, `/admin/prescriptions`, `/admin/products`, `/admin/inventory`, `/admin/manufacturers`, `/admin/customers`, `/admin/promotions`, `/admin/reports`, `/admin/content/*`, `/admin/settings/*` | `/api/admin/dashboard`, `/summary`, `/orders`, `/prescriptions`, `/products`, `/inventory`, `/manufacturers`, `/customers`, `/reports/:key`, `/content/*`, `/settings/*` |

## 3.2 Key Business Rules Confirmed in the Environment

The following configuration was read directly from the running application and forms the expected-result baseline for calculation and rule-based test cases.

| Rule / Configuration | Current Value |
|---|---|
| Home district (inside-city reference) | Dhaka |
| Delivery charge — inside home district | BDT 60 |
| Delivery charge — outside home district | BDT 120 |
| Free delivery threshold | Order value of BDT 2,000 or above |
| Maximum quantity per item | 200 units |
| Enabled payment method | Cash on Delivery (COD) only; bKash and Nagad configured but disabled |
| Delivery coverage | 64 districts of Bangladesh |
| Product departments | Medicines, Health & Wellness, Personal Care, Baby Care, Skin Care, Nutrition |
| Order status lifecycle | Pending → Confirmed → Packed → Shipped → Delivered; Cancelled as terminal exception |
| Prescription review states | Pending review → Approved / Rejected |
| Sales definition (reporting) | MRP value − discount + delivery charge; cancelled orders excluded |
| Order reference format | `NP-1030xx` |

# 4. Test Approach and Test Types

Testing is primarily manual, black-box and risk-based, organised into three execution cycles. Priority is assigned by business impact: revenue-affecting and patient-safety-affecting paths are tested first and most deeply.

## 4.1 Test Types Applied

| Test Type | Purpose and Technique | Applies To |
|---|---|---|
| **Smoke Testing** | Short confidence check after every deployment to confirm the build is testable: site loads, login works, catalogue lists, cart accepts an item, admin dashboard opens. | Both |
| **Functional Testing** | Requirement-by-requirement verification of each module against its expected behaviour, using positive and negative paths. | Both |
| **UI / UX Testing** | Comparison against approved designs: layout, spacing, typography, imagery, iconography, empty states, loading states, error states, Bangla and English text rendering. | Both |
| **Validation & Boundary Testing** | Field-level validation using equivalence partitioning and boundary value analysis — mobile number format, password policy, quantity limits (1, 2, 199, 200, 201), price and stock boundaries, file size and type for uploads. | Both |
| **Calculation Testing** | Independent recalculation of subtotal, discount, delivery charge, free-delivery threshold and grand total, and reconciliation of dashboard and report figures against underlying order data. | Both |
| **Integration Testing** | Verification that a change made in the Admin Dashboard appears correctly and promptly on the Customer Website, and that customer actions appear correctly in the admin. | Cross |
| **Compatibility Testing** | Execution of the core journey across the browser and device matrix in Section 5.3, including responsive breakpoints. | Both |
| **Security Testing** | Application-level checks: role escalation attempts, direct URL and direct API access without a session, session expiry and logout behaviour, IDOR on order and prescription identifiers, input sanitisation for XSS and injection patterns, and file-upload restrictions. | Both |
| **Regression Testing** | Re-execution of an agreed regression suite after defect fixes and before sign-off, to confirm that corrections have not broken working functionality. | Both |

## 4.2 Test Design Techniques

- **Equivalence partitioning** for input domains such as districts, quantities and prices.
- **Boundary value analysis** for the free-delivery threshold (BDT 1,999 / 2,000 / 2,001), maximum quantity per item (199 / 200 / 201) and stock limits (stock − 1 / stock / stock + 1).
- **Decision tables** for delivery-charge determination, which depends on district and order value together.
- **State transition testing** for the order status lifecycle and the prescription review lifecycle, including invalid transitions.
- **Use-case / scenario testing** for the end-to-end journeys in Section 9.
- **Error guessing** based on QA experience with e-commerce and pharmacy platforms.

## 4.3 Test Case Identification

Each test case carries a unique identifier so that defects, evidence and coverage can be traced back to a requirement.

| Format | Example | Meaning |
|---|---|---|
| `NP-<MODULE>-TC-<NNN>` | `NP-CHK-TC-014` | Checkout module (Customer Website), test case 014 |
| `NP-<MODULE>-TC-<NNN>` | `NP-ORM-TC-027` | Order Management module (Admin Dashboard), test case 027 |
| `NP-E2E-TC-<NNN>` | `NP-E2E-TC-003` | Cross-application end-to-end test case 003 |
| `NP-SC-<APP>-<NNN>` | `NP-SC-CW-045` | Test scenario 045, Customer Website (document NP-QA-TS-001) |

# 5. Test Environment and Test Data

## 5.1 Environment Details

| Item | Detail |
|---|---|
| Environment | Test / Staging |
| Application URL | https://202.4.101.118:1165/ |
| Customer Website | Root path of the application |
| Admin Dashboard | `/admin` (role-protected) |
| Protocol | HTTPS on a non-standard port (1165), accessed by IP address |
| Front-end | React single-page application (Vite production build) |
| Back-end | REST API on the same origin under `/api` |
| Session model | Cookie-based session with optional *remember me* |

## 5.2 Test Accounts

| Role | Credential | Purpose |
|---|---|---|
| Administrator | Mobile `01700000000` | Full admin access — all dashboard modules, settings and reports |
| Customer (existing) | To be provisioned by QA | Order history, repeat purchase, saved addresses, wishlist |
| Customer (new) | Created during execution | Registration, first-order and onboarding journeys |
| Guest | No session | Browsing, guest cart behaviour, cart merge on login, access-control checks |

> **Credential handling.** Administrator credentials are shared only with assigned QA personnel and are not recorded in defect reports, screenshots or test evidence. The administrator password must be rotated after the test window closes.

## 5.3 Browser and Device Matrix

| Platform | Browser / Version | Coverage Level |
|---|---|---|
| Windows 11 Desktop | Google Chrome (latest) | Full — all test cases |
| Windows 11 Desktop | Microsoft Edge (latest) | Core journeys and admin suite |
| Windows 11 Desktop | Mozilla Firefox (latest) | Core journeys |
| Android Phone | Chrome for Android | Full — customer website; smoke — admin |
| iPhone | Safari (iOS) | Core customer journeys |
| Tablet | Chrome / Safari | Responsive layout verification |

Responsive breakpoints to be verified: 360 px, 414 px, 768 px, 1024 px, 1366 px and 1920 px.

## 5.4 Test Data Requirements

- Products across all six departments, including in-stock, low-stock, out-of-stock and hidden items.
- At least ten prescription-required products, to exercise the prescription gate.
- Products with and without a discount, and products with and without a photograph.
- Delivery addresses inside Dhaka and in at least five districts outside Dhaka.
- Cart values below, exactly at, and above the BDT 2,000 free-delivery threshold.
- Prescription image files in valid formats, plus oversized and invalid-format files for negative testing.
- Orders pre-seeded in each status: Pending, Confirmed, Packed, Shipped, Delivered and Cancelled.
- Customer accounts with no orders, with a single order, and with a long order history.

- Bangla text, special characters and very long strings for input-field robustness.

# 6. Entry, Exit and Suspension Criteria

## 6.1 Entry Criteria

- The build is deployed to the test environment and the smoke test passes.
- Requirements and design references are available and baselined.
- Test cases are written and peer-reviewed.
- Test accounts and test data are provisioned and verified.
- Defect tracking is configured and accessible to the team.

## 6.2 Exit Criteria

- 100% of planned test cases executed.
- No open **Critical** or **High** severity defects.
- Medium severity defects either resolved or formally accepted by the Project Manager with a documented workaround.
- Test case pass rate of at least **95%**.
- All Critical and High defects retested and closed, with the regression suite re-executed and passing.
- Test Summary Report prepared and signed off.

## 6.3 Suspension and Resumption Criteria

| Testing is suspended when | Testing resumes when |
|---|---|
| The test environment is unavailable or unstable for more than four continuous hours. | The environment is restored and the smoke test passes. |
| A Critical defect blocks a major flow (for example, checkout or admin login fails entirely). | A fix is deployed and verified by smoke test. |
| The smoke test fails on a new build. | A corrected build is delivered and the smoke test passes. |
| More than 30% of executed cases fail in a single cycle, indicating the build is not ready. | The development team delivers a stabilised build. |
| Required test data or credentials are not available. | Data and access are provisioned. |

## 7. Customer Website — Module Test Coverage

Priority key: P1 Critical P2 High P3 Medium

| # | Module | Key Test Scenarios | Pri. | Cases |
|---|---|---|---|---|
| 1 | **Home Page & Merchandising** `CW-HOM` | Page loads within the agreed time; hero slider rotates, navigates and links correctly; promotional top bar displays the correct free-delivery message and resolves placeholders (delivery charge, home district, threshold); department tiles display all six departments and route correctly; featured products and offer sections render with correct price, discount badge and stock state; brand/manufacturer strip shows only manufacturers flagged for the home page; footer links, hotline number and address are correct; empty-state behaviour when a section has no content. | P2 | 8 |
| 2 | **Navigation & Category Browsing** `CW-NAV` | Header, mega-menu and mobile drawer navigation; department-to-category drill-down across all six departments; category listing shows only products of that category; breadcrumb accuracy; pagination and infinite scroll; sorting (price ascending/descending, name, newest); filtering by category, manufacturer, price range, discount and stock availability; multiple filters combined; filter reset; result count accuracy; behaviour when a filter returns no results; browser back and forward preserve filter state. | P2 | 10 |
| 3 | **Search & Autocomplete** `CW-SRC` | Search by brand name, generic name, manufacturer and partial keyword; autocomplete suggestions appear, are relevant, and show name, generic, strength, formation, price and stock; selecting a suggestion opens the correct product; case-insensitive and whitespace-tolerant matching; Bangla input; special characters and SQL/script-like strings are handled safely; no-result message and suggested alternatives; very long query strings; rapid typing does not produce stale or duplicated results; search result count matches the listing. | P1 | 10 |
| 4 | **Product Detail** `CW-PDP` | Correct display of name, generic, strength, formation, pack unit, manufacturer, category, MRP, selling price, discount percentage and saving; product image or fallback artwork; in-stock, low-stock and out-of-stock indicators; prescription-required badge is shown on controlled medicines; quantity selector honours minimum 1 and the per-item maximum; add to cart and add to wishlist; related and same-generic product suggestions; direct URL by slug; invalid or deleted slug returns a proper not-found page; out-of-stock product cannot be added to cart. | P1 | 10 |
| 5 | **Cart** `CW-CRT` | Add single and multiple items; increase and decrease quantity; remove a line; clear the cart; cart badge count accuracy; subtotal recalculation on every change; per-item maximum of 200 enforced and a clear message shown at the limit; quantity cannot exceed available stock; guest cart persists across page reloads; guest cart merges correctly into the account cart on login without duplication or loss; price changes made in the admin are reflected in the cart; a product hidden or deleted in the admin is handled gracefully; empty-cart state and its call to action; delivery charge preview and free-delivery progress indicator. | P1 | 14 |
| 6 | **Wishlist** `CW-WSH` | Add and remove items from product cards, product detail and the wishlist page; wishlist icon state reflects saved items; wishlist persists across sessions; move an item from wishlist to cart; out-of-stock item behaviour in the wishlist; empty-wishlist state; wishlist requires authentication, and the intended action resumes after login. | P3 | 6 |
| 7 | **Registration** `CW-REG` | Successful registration with valid details; mandatory field validation; Bangladeshi mobile number format validation (11 digits, valid operator prefix); duplicate mobile number is rejected with a clear message; email format validation where provided; password policy enforcement (length and complexity); confirm-password mismatch; terms acceptance where required; automatic login after registration; the new account appears immediately in the admin customer list; submission is protected against double-click duplicate creation. | P1 | 12 |
| 8 | **Login, Logout & Session** `CW-LOG` | Login with valid credentials; invalid password and unregistered mobile produce a generic, non-enumerating error; empty-field validation; *remember me* keeps the session across browser restarts; without it, the session ends as designed; redirect to the originally requested page after login; logout clears the session and protected pages become inaccessible; using the browser back button after logout does not expose account data; session expiry is handled with a clean redirect rather than an error; repeated failed attempts are throttled or locked out. | P1 | 12 |

| | | | | |
|---|---|---|---|---|
| 9 | **Password Recovery** `CW-FGT` | Available recovery channels are listed correctly; reset code or link is sent to the registered mobile or email; unregistered number yields a non-enumerating response; code validation, expiry and single use; incorrect code is rejected; new password honours the password policy; login succeeds with the new password and fails with the old one; resend throttling; reset link cannot be replayed after use. | P2 | 8 |
| 10 | **Account & Profile** `CW-ACC` | Account summary shows accurate order count, spend and recent activity; view and update profile (name, email); mobile number change rules; change password with current-password verification and policy enforcement; address book — add, edit, delete and set default; address fields including recipient name, phone, address line, area and district; district dropdown lists all 64 districts; the default address is preselected at checkout; the last remaining address cannot be deleted while required; validation on all address fields. | P2 | 12 |
| 11 | **Prescription Upload** `CW-RXU` | Upload a prescription image from the account area and during checkout; accepted file formats and rejection of invalid types; maximum file size enforced with a clear message; multiple files where supported; uploaded prescription is listed with its status; status progresses through Pending review, Approved and Rejected, with the rejection reason visible; an approved prescription can be attached to an order at checkout; a pending prescription cannot be used to release a controlled medicine; the upload appears immediately in the admin prescription queue; the image is legible at full size; the file is not publicly accessible without authorisation. | P1 | 10 |
| 12 | **Checkout & Order Placement** `CW-CHK` | Checkout requires authentication; saved addresses load and a new address can be added inline; district selection drives the delivery charge — BDT 60 inside Dhaka, BDT 120 elsewhere; free delivery applies at BDT 2,000 and above, verified at 1,999, 2,000 and 2,001; subtotal, discount, delivery charge and grand total are arithmetically correct; Cash on Delivery is selectable and disabled wallet methods are not; prescription attachment for controlled items is mandatory and enforced server-side; customer note is captured; order placement succeeds and the order confirmation page shows the correct order number and details; stock is decremented correctly; the order appears in the admin order list immediately; placing an order with an item that went out of stock mid-checkout is handled gracefully; double submission does not create two orders; interrupting the network during submission does not corrupt the order. | P1 | 18 |
| 13 | **Order History & Tracking** `CW-ORD` | Order list shows all orders of the logged-in customer in correct order; order detail shows items, quantities, unit price, line totals, delivery charge, grand total, payment method and payment status; status timeline reflects the current status and history; status changes made by the admin appear on the customer side; cancellation is permitted only in allowed statuses and blocked afterwards; reorder where supported; invoice or receipt view; a customer cannot access another customer's order by altering the order identifier in the URL; empty-order state. | P1 | 10 |
| 14 | **Content Pages & Newsletter** `CW-CNT` | Static pages render from admin-managed content; terms, privacy, delivery information, help and FAQ pages are reachable and correct; FAQ expand and collapse; manufacturer listing page and per-manufacturer product listing; newsletter subscription with email validation, duplicate handling and confirmation message; contact and hotline details match store settings; broken-link check across the site. | P3 | 9 |

**Customer Website total: 149 test cases**

## 8. Admin Dashboard — Module Test Coverage

| # | Module | Key Test Scenarios | Pri. | Cases |
|---|---|---|---|---|
| 1 | **Admin Access & Role Gate** `AD-SEC` | Administrator logs in successfully and lands on the dashboard; a customer-role account is denied access to `/admin` and to admin API endpoints; an unauthenticated visitor is redirected to login rather than shown admin content; direct entry of deep admin URLs without a session is blocked; admin session expiry and logout; the back button after logout does not reveal cached admin data; the admin area is not reachable through client-side manipulation; last login timestamp is recorded and displayed. | P1 | 8 |
| 2 | **Dashboard & Analytics** `AD-DSH` | Attention counters (orders to confirm, orders in progress, prescriptions to review, low-stock items) are accurate against the underlying data; today's orders and today's sales are correct; period comparison against the previous period calculates correctly; the seven-day trend chart plots sales, orders, customers, prescriptions and products added accurately; order status breakdown totals reconcile with the order list; top-selling products list shows correct quantity and revenue; date-range switching refreshes all tiles consistently; zero-data and single-data-point days render without error; figures reconcile with the Reports module for the same period. | P1 | 8 |
| 3 | **Order Management** `AD-ORD` | Order list displays order number, date, customer, phone, district, item and piece counts, total, payment method, payment status, order status and prescription flag; search by order number, customer name and phone; filter by status, date range and district; pagination and sorting; order detail shows the full item breakdown, pricing, delivery charge, totals, delivery address and customer note; status progression Pending → Confirmed → Packed → Shipped → Delivered; cancellation with reason; invalid or backward transitions are prevented; every status change is written to the order history with the operator name and timestamp; payment status update; printing or exporting an invoice; the customer sees the updated status; a cancelled order is excluded from sales figures; stock is restored on cancellation where applicable; an order containing a prescription item cannot be advanced before the prescription is approved. | P1 | 14 |
| 4 | **Prescription Review** `AD-RXA` | Pending prescriptions appear in the review queue immediately after a customer uploads; the prescription image opens, zooms and is legible; approve and reject actions with a mandatory reason on rejection; status counters for Approved and Rejected update correctly; the linked order and customer are visible from the prescription record; the customer is notified of the outcome and sees the updated status; an approved prescription can be used at checkout and a rejected one cannot; filtering by status and searching by customer; the audit record captures the reviewing pharmacist and the time; prescription files are served only to authorised sessions. | P1 | 8 |
| 5 | **Product Catalogue** `AD-PRD` | Create a product with all attributes — name, generic, strength, formation, pack unit, manufacturer, category, MRP, selling price, discount, stock, prescription flag and featured flag; mandatory field validation; selling price cannot exceed MRP; discount percentage calculates and displays correctly; slug generation and uniqueness; product image upload, replacement and removal, including format and size limits; edit and update an existing product; hide and unhide; delete and its effect on existing orders and carts; a newly created product appears on the customer site with correct data; search and filter within the product list; bulk actions where supported. | P1 | 12 |
| 6 | **Inventory & Stock** `AD-INV` | Inventory list shows stock on hand, low-stock flag, hidden state, photo presence and POS sales figures; stock update for a single product and in bulk; low-stock threshold behaviour and the resulting dashboard counter; quick filters for low stock, out of stock, hidden, missing photo and missing data; setting stock to zero marks the product out of stock on the customer site; increasing stock restores availability; stock decrements when an order is placed; negative stock cannot be entered; the maximum purchasable quantity respects both the per-item cap and available stock; concurrent stock edits are handled safely. | P1 | 9 |
| 7 | **Manufacturers / Brands** `AD-MFR` | Manufacturer list with product count and in-stock count; create, edit and delete a manufacturer; duplicate name prevention; logo upload, replacement and removal with format and size validation; the home-page display toggle controls the brand strip on the customer site; deleting a manufacturer that has products is prevented or handled with a clear rule; product counts reconcile with the catalogue; search and sorting. | P3 | 5 |

| | | | | |
|---|---|---|---|---|
| 8 | **Customer Management** `AD-CUS` | Customer list shows name, phone, email, role, active state, registration date, last login, order count and lifetime spend; search by name, phone and email; filter by active state and registration date; customer detail shows order history and addresses; activate and deactivate an account, and a deactivated customer cannot log in; order count and spend reconcile with the order records; a newly registered customer appears immediately; pagination and sorting; personal data is displayed only to authorised roles. | P2 | 7 |
| 9 | **Promotions** `AD-PRM` | Create, edit, activate, deactivate and delete a promotion; discount value and percentage validation; start and end date validation, including an end date earlier than the start date; an active promotion applies on the customer site and is reflected in cart and checkout totals; an expired or deactivated promotion no longer applies; overlapping promotions resolve by the defined rule; promotional pricing is recorded correctly on the order and in reports. | P2 | 6 |
| 10 | **Reports** `AD-RPT` | Each report generates with correct columns and rows — sales summary, orders, products, categories, manufacturers and brands, customers, delivery areas, delivery performance, prescriptions, and order-state reports for pending, processing, delivered and cancelled; date-range presets (last week, last month) and custom ranges; grouping by day and by month; the sales definition (MRP value − discount + delivery charge, excluding cancelled orders) is applied correctly; totals and averages recompute correctly; report figures reconcile with the dashboard and with the order list for the same period; empty-range behaviour; large-range performance; export to file where available, with correct encoding of Bangla text and currency; drill-through links open the correct records. | P1 | 10 |
| 11 | **Website Content Management** `AD-CNT` | Home page content — top bar message, tagline, hero slides with eyebrow, title, highlight, body text, button labels and links, artwork, tone and statistics toggle; placeholder tokens for delivery charge, home district and free-delivery threshold resolve to the current settings on the customer site; add, reorder, edit and remove hero slides; static page management with rich text; FAQ create, edit, reorder and delete; image library upload, replacement and deletion; every change is reflected on the customer site after save; invalid link handling; unsaved-change warning. | P2 | 7 |
| 12 | **Store Settings** `AD-SET` | **Details:** store name, hotline, hotline hours, email and address save correctly and appear in the site header, footer and contact page. **Delivery:** home district, inside and outside delivery charges, free-delivery threshold and maximum quantity per item save correctly and immediately change checkout behaviour; negative and non-numeric values are rejected. **Payments:** enabling and disabling Cash on Delivery, bKash and Nagad changes the methods offered at checkout; a wallet number is mandatory when its method is enabled; disabling every method is prevented. **Social:** Facebook, Instagram, YouTube and LinkedIn links save, validate as URLs and render in the footer. Each settings group carries a version stamp with the last editor and timestamp — verify that a stale concurrent edit is detected and rejected rather than silently overwriting another operator's change. | P1 | 10 |
| 13 | **Admin UI & Usability** `AD-UIX` | Sidebar navigation and active-state highlighting across all modules; breadcrumbs; table sorting, pagination and page-size selection; loading, empty and error states; success and failure toast messages; confirmation dialogs on destructive actions; unsaved-change protection; keyboard accessibility of primary forms; responsive behaviour on tablet and mobile; consistent date, time and currency formatting throughout; correct rendering of Bangla text in names and addresses. | P3 | 6 |

**Admin Dashboard total: 110 test cases**

## 9. High-Priority End-to-End Test Scenarios

These scenarios cross both applications and represent the highest business risk. They are executed in every test cycle and form the core of the regression suite.

| ID | Scenario | Steps | Expected Result |
|---|---|---|---|
| `NP-E2E-001` | New customer, first order, Cash on Delivery, inside Dhaka | Register a new customer → search a non-prescription product → open product detail → add to cart → open cart → proceed to checkout → add a Dhaka address → select Cash on Delivery → place the order. | Order is created with delivery charge BDT 60; the grand total equals subtotal − discount + 60; the order number follows the `NP-xxxxxx` format; the order appears in the customer's order list and in the admin order list in Pending status; stock is reduced by the ordered quantity. |
| `NP-E2E-002` | Free-delivery threshold boundary | Build a cart of BDT 1,999, observe the total → increase the cart to exactly BDT 2,000 → increase to BDT 2,001 → place the order at each value for a Dhaka and a non-Dhaka address. | At BDT 1,999 the delivery charge applies (60 inside Dhaka, 120 outside); at BDT 2,000 and BDT 2,001 delivery is free and the grand total equals the subtotal. The same rule is applied consistently in the cart preview, at checkout and on the placed order. |
| `NP-E2E-003` | Prescription medicine, full approval cycle | Log in as a customer → add a prescription-required product to the cart → attempt checkout without a prescription → upload a prescription → attempt checkout while the prescription is still pending → log in as administrator → review and approve the prescription → return to the customer account and complete checkout. | Checkout is blocked without a prescription and while the prescription is pending, with a clear message. After approval, the customer's prescription status updates, checkout is permitted, and the order records the linked prescription. A rejected prescription keeps checkout blocked and shows the rejection reason. |
| `NP-E2E-004` | Order lifecycle through to delivery | Place an order as a customer → as administrator move it to Confirmed, then Packed, then Shipped, then Delivered → check the customer view after each change. | Each transition is accepted in the correct sequence and rejected out of sequence; the order history records every change with operator and timestamp; the customer sees the current status and timeline; the dashboard status breakdown and the reports update accordingly. |
| `NP-E2E-005` | Order cancellation and its financial effect | Place an order → cancel it from the customer side while permitted → place a second order and cancel it from the admin side with a reason → review the dashboard and the sales report. | Cancellation is allowed only in permitted statuses; the reason is recorded; cancelled orders are excluded from sales totals in both the dashboard and the sales report; stock is restored where the rule requires it; the customer cannot cancel after the permitted stage. |
| `NP-E2E-006` | Admin change propagates to the storefront | As administrator change a product price, apply a discount, set stock to zero, and then hide the product → after each change check the customer website listing, product detail and cart. | Each change is reflected on the customer site without a stale cache; the discounted price and saving display correctly; a zero-stock product shows as out of stock and cannot be added to the cart; a hidden product is no longer listed and its direct URL returns a proper not-found response; an item already in a cart is handled gracefully. |

| | | | |
|---|---|---|---|
| `NP-E2E-007` | Delivery settings change applied end to end | As administrator change the inside and outside delivery charges and the free-delivery threshold → as a customer build a cart and go to checkout. | The new charges and threshold apply immediately at checkout and in the cart preview; the home page promotional message reflects the new values through its placeholder tokens; the settings record stores the new version, editor and timestamp. |
| `NP-E2E-008` | Guest cart merge on login | As a guest add three products to the cart → log in to an account that already holds two different products, one of which overlaps with the guest cart. | After login the cart contains the correct union of items; the overlapping item has the correct combined quantity without exceeding the per-item maximum; no item is lost or duplicated; the subtotal is recalculated correctly. |
| `NP-E2E-009` | Authorisation boundary | While logged in as a customer, attempt to open admin URLs directly and call admin API endpoints; while logged out, repeat the same attempts; attempt to open another customer's order and prescription by changing the identifier in the URL. | Every attempt is denied with an authorisation or not-found response; no admin data, order data or prescription image is returned; the customer is redirected to login where appropriate; the attempts are not treated as valid sessions. |
| `NP-E2E-010` | Reporting reconciliation | Place a known set of orders across several days and statuses → open the dashboard for the matching period → open the sales, orders and products reports for the same period → compare against a manual calculation. | Order counts, unit counts, discounts, delivery charges, sales totals and average order value agree across the dashboard, all reports and the manual calculation; cancelled orders are excluded consistently; day and month groupings produce the same period totals. |

# 10. Non-Functional Testing

## 10.1 Performance (Baseline)

| Measure | Target |
|---|---|
| Home page first meaningful render | Within 3 seconds on a broadband connection |
| Product listing and search results | Within 2 seconds |
| Autocomplete suggestion response | Within 500 milliseconds |
| Add to cart and cart update | Within 1 second |
| Order placement | Within 5 seconds |
| Admin list screens (orders, products, customers) | Within 3 seconds for a standard page size |
| Report generation for a one-month range | Within 10 seconds |
| Concurrent user baseline | 50 concurrent customer sessions without functional degradation |

## 10.2 Security Checks (Application Level)

- Role escalation: a customer-role session must not reach any admin route or admin API endpoint.
- Unauthenticated access: every protected route and endpoint must deny access without a valid session.
- Insecure direct object reference: order, prescription and address identifiers must not expose another user's records when altered in the URL.
- Session handling: logout invalidates the session; expiry is enforced; session data is not exposed to client-side script inspection.
- Input sanitisation: script and injection patterns submitted into search, name, address, note and content fields must be stored and rendered safely.
- File upload: prescription and product image uploads must restrict file type and size, and uploaded files must not be executable or publicly enumerable.
- Credential handling: passwords must never appear in responses, URLs, logs or client storage; login errors must not reveal whether an account exists.
- Transport: the application must be served over HTTPS with no mixed content.

## 10.3 Usability and Accessibility

- Consistent terminology, currency formatting (BDT) and date formatting across both applications.
- Clear, actionable error messages in place of technical error text.
- Correct rendering of Bangla and English text, including in names, addresses and content pages.
- Keyboard navigation and visible focus states on primary forms.
- Sufficient colour contrast for text, badges and status indicators.
- Meaningful alternative text on product and banner images.
- Touch targets of adequate size on mobile.

# 11. Defect Management

## 11.1 Severity Definitions

| Severity | Definition | Examples |
|---|---|---|
| Critical | Core business function is unusable, data is corrupted, money is calculated wrongly, or patient safety or security is compromised. No workaround. | Checkout fails; order total is wrong; a customer can reach admin functions; a prescription medicine can be purchased without approval; stock is not decremented. |
| High | A major feature does not work as specified; a workaround exists but is impractical for normal use. | Order status cannot be changed; search returns incorrect results; prescription cannot be uploaded; a report shows incorrect figures. |
| Medium | A feature works incorrectly in a limited way, or a non-critical function fails, with an acceptable workaround. | Filter resets unexpectedly; pagination skips a page; validation message is unclear; sorting order is wrong. |
| **Low** | Cosmetic or minor issue with no functional impact. | Spelling or label inconsistency; minor alignment or spacing deviation; inconsistent icon. |

## 11.2 Priority Definitions

| Priority | Expected Resolution |
|---|---|
| P1 — Immediate | Fix within 24 hours; blocks further testing or release |
| P2 — High | Fix within the current test cycle |
| P3 — Medium | Fix before release sign-off |
| P4 — Low | May be deferred to a subsequent release with approval |

## 11.3 Defect Lifecycle

New → Assigned → In Progress → Fixed → Retest → Closed, with Reopened and Deferred / Rejected as alternate outcomes. A defect may be closed only by the QA engineer who verified the fix.

## 11.4 Defect Report Contents

- Defect identifier, title and the related test case identifier.
- Application (Customer Website or Admin Dashboard), module and environment.
- Severity and priority.
- Precise steps to reproduce, with the test data used.
- Expected result and actual result.
- Screenshot or screen recording, and any relevant browser console or network detail.
- Browser, device and operating system; date and reporter.

## 12. Test Schedule and Effort Estimation

| Phase | Activity | Duration | Resource | Output |
|---|---|---|---|---|
| 1 | Test planning and requirement study | 2 working days | QA Lead | Approved test plan |
| 2 | Test case design and peer review | 5 working days | QA Lead + 2 Engineers | Reviewed test case repository |
| 3 | Environment setup and test data preparation | 1 working day | QA Engineer + DevOps | Ready environment and seeded data |
| 4 | Cycle 1 — full functional execution | 6 working days | 2 QA Engineers | Cycle 1 execution report and defect log |
| 5 | Defect fixing window (development) | 3 working days | Development Team | Corrected build |
| 6 | Cycle 2 — retest and regression | 4 working days | 2 QA Engineers | Cycle 2 report and updated defect status |
| 7 | Non-functional testing (compatibility, performance, security) | 3 working days | QA Engineer | Non-functional findings report |
| 8 | Cycle 3 — final regression and sanity | 2 working days | QA Team | Final regression result |
| 9 | Test summary report and sign-off | 1 working day | QA Lead | Test Summary Report and release recommendation |
| | **Total** | **27 working days — approximately 4 calendar weeks (excluding the development fixing window overlap)** | | |

## 13. Roles and Responsibilities

| Role | Responsibility |
|---|---|
| QA Lead | Owns this test plan; designs and reviews test cases; allocates work; manages defect triage; prepares the test summary report; recommends release readiness. |
| QA Engineers (2) | Execute test cases; record results and evidence; raise and retest defects; perform regression and compatibility runs. |
| Development Team | Delivers builds to the test environment; analyses and fixes defects; provides technical clarification; confirms fixes with release notes. |
| Project Manager | Approves the plan and the schedule; arbitrates scope and priority; accepts or defers Medium and Low defects; approves release. |
| Business / Pharmacy Owner | Confirms business rules, pricing, delivery policy and prescription handling requirements; performs user acceptance testing. |
| DevOps / System Administrator | Maintains the test environment; performs deployments; restores data and provides access. |

## 14. Risks and Mitigation

| Risk | Impact | Mitigation |
|---|---|---|
| Test environment instability or unannounced deployments during execution | High | Agree a deployment freeze during execution windows; require release notes for every build; run a smoke test before each cycle. |
| Incomplete or changing requirements | High | Baseline requirements before Cycle 1; route later changes through change control with a schedule impact assessment. |
| Insufficient or unrealistic test data, particularly prescription products and multi-district addresses | Medium | Prepare and seed the data set listed in Section 5.4 before execution; keep a restorable snapshot. |
| Payment gateways (bKash, Nagad) are currently disabled and cannot be fully validated | Medium | Test the configuration and selection behaviour now; schedule a separate gateway certification cycle before those channels are switched on in production. |
| Large catalogue (several thousand products) may expose performance issues only at scale | Medium | Execute listing, search and report tests against a production-sized data volume rather than a small sample. |
| Limited device coverage for iOS and older Android versions | Medium | Supplement physical devices with an emulator or a cloud device service for the core journeys. |
| Compressed timeline reduces regression depth | Medium | Maintain a prioritised regression suite so that P1 and P2 coverage is protected even if the schedule contracts. |
| Key QA resource unavailability | Low | Keep test cases detailed enough for another engineer to execute; cross-train across modules. |

## 15. Test Deliverables

| Deliverable | Owner | When |
|---|---|---|
| Test Plan (this document) | QA Lead | Before execution begins |
| Test Case Repository | QA Lead / Engineers | End of design phase |
| Requirement Traceability Matrix | QA Lead | End of design phase |
| Daily Execution Status | QA Engineers | Daily during execution |
| Defect Log and Triage Report | QA Lead | Continuous |
| Cycle Execution Reports (1, 2, 3) | QA Lead | End of each cycle |
| Compatibility and Non-Functional Report | QA Engineer | End of Phase 7 |
| Test Summary Report and Release Recommendation | QA Lead | End of the engagement |
| Test Evidence Archive (screenshots, recordings) | QA Engineers | Continuous |

## 16. Observations from Environment Assessment

The following points were noted while assessing the live test environment on 21 October 2026. They are not defects, but each carries a testing consequence that has been folded into the plan above.

| Observation | Testing Consequence |
|---|---|
| The application is reachable by IP address on port 1165 over HTTPS, which produces a certificate trust warning in browsers. | Testers must note that this warning is an environment artefact, not a product defect. Certificate and domain configuration must be re-verified on the production domain before release, and mixed-content checks repeated there. |
| The API rejects requests that do not carry the application's expected client header, returning *Request rejected* with HTTP 400. | This is a positive control. Security test cases should confirm that this check cannot be used to bypass authentication, and that the resulting error message is handled gracefully in the user interface rather than surfacing raw API text. |
| Store settings are versioned, recording a version number, the last editor and the last update time per settings group. | Concurrency test cases are required: two administrators editing the same settings group must not silently overwrite one another. This is covered in module `AD-SET`. |
| Only Cash on Delivery is currently enabled; bKash and Nagad exist in the configuration but are switched off. | Wallet payment flows cannot be fully validated in this cycle. Configuration and checkout-visibility behaviour is tested now; a separate gateway certification cycle is required before enabling them. |
| The environment reports a large number of low-stock items and a substantial catalogue across many manufacturers. | Inventory, listing, search and report test cases should be executed against this production-sized volume, since pagination and performance issues typically appear only at scale. |
| Existing order and customer records include entries created by earlier automated test runs. | Test data must be clearly segregated and identifiable so that reporting reconciliation in `NP-E2E-010` is performed against a known, controlled data set. |
| The order history records the operator name and timestamp for each status change. | Audit-trail verification is included in module `AD-ORD`; every status transition must produce a complete and accurate history entry. |

## 17. Approval

By signing below, the parties confirm that the scope, approach, schedule and exit criteria defined in this test plan are understood and accepted.

| Role | Name | Signature | Date |
|---|---|---|---|
| Prepared by — QA Lead | Md. Masud Rana | | |
| Reviewed by — Project Manager | | | |
| Reviewed by — Development Lead | | | |
| Approved by — Business Owner | | | |